

OrientationJ in the scientific literature

304 publications, 2013–2026, using the ImageJ/Fiji plugin OrientationJ (D. Sage, Biomedical Imaging Group, EPFL) — with the sentence from each paper describing how it was used.

194 fields of application · 61 distinct plugin modes · article titles and author lists are in OrientationJ_References_raw.csv · resources and primary references: OrientationJ_Appendices.pdf · compiled with AI assistance, 7 August 2026

#	Year	OrientationJ mode	Field / application	Reference	Journal	DOI	Usage extract
1	2013	Coherency	collagen/ECM in tumors	Perry et al. (2013)	BMC Cancer	10.1186/1471-2407-13-411	<i>the coherency parameter was calculated on these same collagen I IF images using the ImageJ plugin OrientationJ</i>
2	2014	Color-coded orientation map	microtubule cytoskeleton, planar cell polarity	Matis et al. (2014)	eLife	10.7554/elife.02893	<i>Orientations of MTs, in the regions shown, calculated using OrientationJ, aligned with the P-D axis at each stage and location (arrows).</i>
3	2014	Coherency and orientation analysis	plant tissue regeneration and cell organization	Zajączkowska (2014)	Planta	10.1007/s00425-014-2142-y	<i>Cell ordering within the overgrowth tissue was estimated using the OrientationJ image analysis plugin based on structure tensor.</i>
4	2014	Distribution (directionality histogram)	vascular endothelial cell alignment	Li et al. (2014)	Nature	10.1038/nature13701	<i>Automated quantification of cell orientation relative to the direction of shear stress was determined using OrientationJ software, also an ImageJ plugin</i>
5	2015	orientation calculation	cardiac tissue engineering	Pasqualini et al. (2015)	Stem Cell Reports	10.1016/j.stemcr.2015.01.020	<i>OrientationJ (Rezakhaniha et al., 2012) plugins were used to highlight the filamentous structure and calculate the orientations of foreground pixels, respectively.</i>
6	2015	Coherency map	cell biophysics / gold nanoparticles	Kalies et al. (2015)	PLOS ONE	10.1371/journal.pone.0124052	<i>This was done using the ImageJ plugin OrientationJ [38 , 39], that allows to measure the coherency of local image features.</i>
7	2015	Distribution (directionality histogram)	cellular and subcellular oriented structures	Kartasalo et al. (2015)	BMC Bioinformatics	10.1186/s12859-015-0782-y	<i>Previously published general purpose tools include the ImageJ plug-ins FibrilTool and OrientationJ, which allow the quantification of orientation distributions based on intensity gradients.</i>
8	2015	Coherency map	collagen/ECM in tumors	Lee et al. (2015)	IntraVital	10.1080/21659087.2015.1055431	<i>Collagen coherency and amount was calculated from the left and right images via the OrientationJ plugin for ImageJ and thresholded area respectively.</i>
9	2015	orientation/coherency analysis	collagen/ECM materials	Lin et al. (2015)	Materials	10.3390/ma8020551	<i>The orientation of collagen fiber network was analyzed using the OrientationJ plugin [20] in ImageJ software (NIH, Bethesda, MD, USA).</i>
10	2015	structure-tensor order parameter	fetal white matter tracts / neuronal fiber organization	Mitter et al. (2015)	Frontiers in Neuroanatomy	10.3389/fnana.2015.00164	<i>ST analysis was performed with the OrientationJ plugin for ImageJ using a cubic spline gradient with a Gaussian window of 5 pix.</i>
11	2015	Distribution (directionality histogram)	materials/nanofibers	Hotaling et al. (2015)	Data in Brief	10.1016/j.dib.2015.07.012	<i>The file labeled " Figure 5 C – Orientation Histogram.xlsx" contains the frequency of occurrence of a given fiber orientation as defined by the OrientationJ plugin.</i>
12	2015	Distribution (directionality histogram)	materials/nanofibers	Jenkins et al. (2015)	Nanoscale Research Letters	10.1186/s11671-015-0972-6	<i>OrientationJ, a plug-in for ImageJ, was used to process SEM images and generate a table of pixels counted versus degree of orientation.</i>
13	2015	orientation/coherency analysis	materials/nanofibers	Bhadra et al. (2015)	Scientific Reports	10.1038/srep16817	<i>SEM images were obtained using the ImageJ ® 1.48 package with DiameterJ and OrientationJ plugins.</i>
14	2015	orientation/coherency analysis	scaffold tissue engineering	Zhang et al. (2015)	Science Advances	10.1126/sciadv.1500423	<i>Orientation of the scaffold struts was quantified using an ImageJ plug-in, OrientationJ</i>
15	2015	Dominant Direction	tendon/mineralized tissue biomechanics	Breidenbach et al. (2015)	Journal of Orthopaedic Research	10.1002/jor.22899	<i>Collagen fiber orientation of the mineralized and unmineralized insertion was calculated using the FIJI plug-in OrientationJ</i>
16	2016	Measure	actin cytoskeleton	Nesmith et al. (2016)	Journal of Cell Biology	10.1083/jcb.201603111	<i>the orientation of actin fibers was determined using the OrientationJ plugin (Rezakhaniha et al., 2012), and the actin OOP was calculated for each sample.</i>
17	2016	Distribution (directionality histogram)	actin cytoskeleton	Squarr et al. (2016)	Journal of Cell Biology	10.1083/jcb.201508081	<i>Rose plots show density distribution of local angles of actin bundles (calculated using OrientationJ).</i>
18	2016	Dominant Direction	cardiac tissue engineering	Laksari et al. (2016)	Royal Society Open Science	10.1098/rsos.160365	<i>OrientationJ determines the fibre directions in each area of interest by defining a weighted inner product and calculating the eigenvectors of the structure tensor</i>
19	2016	Color-coded orientation map	collagen/ECM in breast tissue	Sherratt et al. (2016)	Breast Cancer Research	10.1186/s13058-016-0701-9	<i>Polarised light microscopy images of PSR-stained sections of c low and d high MD breast processed using the Image J plugin, OrientationJ.</i>
20	2016	Coherency map	collagen/ECM in breast tissue	McConnell et al. (2016)	Breast Cancer Research	10.1186/s13058-015-0664-2	<i>Polarised light microscopy of PSR-stained sections assessed using OrientationJ: e low MD, f high MD.</i>
21	2016	orientation/coherency analysis	collagen/ECM in tumors	Kopanska et al. (2016)	PLOS ONE	10.1371/journal.pone.0156442	<i>Collagen orientation was measured using "OrientationJ" plugin of ImageJ [56].</i>
22	2016	Measure	microbial mat architecture	Chan et al. (2016)	Frontiers in Microbiology	10.3389/fmicb.2016.00796	<i>Directionality measurements were taken using the ImageJ 2.0.0 OrientationJ Measure plugin</i>
23	2016	orientation/coherency analysis	neuronal alignment in developing brain	Koser et al. (2016)	Nature Neuroscience	10.1038/nn.4394	<i>These were analyzed using the OrientationJ plugin.</i>
24	2016	Distribution (directionality histogram)	vascular tissue mechanics	Bell et al. (2016)	American Journal of Physiology-Heart and Circulatory Physiology	10.1152/ajpheart.00002.2016	<i>The OrientationJ Distribution plugin for Fiji (37) was used to quantify the orientation of the intimal and adventitial elastic fiber networks, and the collagen network.</i>
25	2017	orientation and coherency analysis	actin cytoskeleton	Fritzsche et al. (2017)	Science Advances	10.1126/sciadv.1603032	<i>The orientation and isotropy properties of the actin filaments in a given region of an (experimental or simulated) image were computed based on the evaluation of the structure tensor in a local neighborhood using the Java plugin for ImageJ (http://imagej.nih.gov/) "OrientationJ.</i>
26	2017	Color-coded orientation map	actin cytoskeleton	Wirshing et al. (2017)	Molecular Biology of the Cell	10.1091/mbc.e17-01-0029	<i>OrientationJ was used to false-color actin bundles according to their orientation in selected cells indicated by yellow boxes in A.</i>
27	2017	orientation/coherency analysis	actin cytoskeleton	Horejs et al. (2017)	Nature Communications	10.1038/ncomms15509	<i>Actin fibre orientation was evaluated using the OrientationJ plug-in for ImageJ</i>
28	2017	Measure	actin cytoskeleton	Fritzsche et al. (2017)	Nature Communications	10.1038/ncomms14347	<i>The orientation and isotropy properties of a given region of interest in an image were computed based on the evaluation of the structure tensor in a local neighbourhood using the Java plug-in for ImageJ 'OrientationJ'.</i>
29	2017	orientation/coherency analysis	cardiac muscle myofibrils	Panzer et al. (2017)	Scientific Reports	10.1038/srep46438	<i>The orientation of myofibers was analyzed using ImageJ and an ImageJ plugin, OrientationJ</i>
30	2017	orientation/coherency analysis	collagen/ECM in tumors	Roy et al. (2017)	Laboratory Investigation	10.1038/labinvest.2016.146	<i>ECM fiber orientation analysis Monochromatic images corresponding to fibronectin fiber acquisitions were analyzed via ImageJ's OrientationJ plugin</i>
31	2017	Color-coded orientation map	collagen/ECM in tumors	Franco-Barraza et al. (2017)	eLife	10.7554/elife.20600	<i>ECM fiber angle distributions, measured with Image-J's 'OrientationJ' plug, are represented by the various colors</i>
32	2017	Coherency map	corneal epithelial tissue engineering	Lombardo et al. (2017)	Scientific Reports	10.1038/s41598-017-05486-8	<i>The plugin OrientationJ in ImageJ was used for analysis of stromal collagen fiber orientations, based on the evaluation of the structure tensor in a local neighborhood</i>
33	2017	Measure	corneal tissue engineering, collagen organization	Gouveia et al. (2017)	Biomaterials	10.1016/j.biomaterials.2016.12.010	<i>Data was analyzed using the OrientationJ plugin from ImageJ v1.46 for measuring collagen fibril dimensions, orientation, and distribution.</i>

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34	2017	orientation/coherency analysis	electrospun nanofiber scaffolds	Chen et al. (2017)	ACS Applied Materials & Interfaces	10.1021/acsami.7b07151	<i>The plugin OrientationJ was employed for this purpose, and the images were analyzed over the same ROI.</i>
35	2017	Distribution (directionality histogram)	endothelial cell alignment in tumors	Webster et al. (2017)	Oncotarget	10.18632/oncotarget.15039	<i>Quantification of cell orientation relative to the direction of shear stress was determined using OrientationJ software</i>
36	2017	structure tensor eigenvector direction	epithelial tissue organization	Saw et al. (2017)	Nature	10.1038/nature21718	<i>OrientationJ was used to detect the direction of largest eigenvector of the structure tensor of the image for each pixel</i>
37	2017	Vector Field	extracellular matrix/tissue regeneration	Marrero et al. (2017)	Regeneration	10.1002/reg2.75	<i>measuring vectors in 32-bit thresholded channels of ER-TR7 + staining using the OrientationJ plugin</i>
38	2017	Distribution (directionality histogram)	materials/laser-induced surface structures	Gnilitskyi et al. (2017)	Scientific Reports	10.1038/s41598-017-08788-z	<i>different spectra (b) of orientations θ can be obtained with the module OrientationJ 23 of the open software ImageJ 59</i>
39	2017	Coherency map	materials/nanofibers	Modarres et al. (2017)	Scientific Reports	10.1038/s41598-017-13565-z	<i>The scores for the aligned category by image classification are compared with a normalised value for alignment (coherency) using the OrientationJ plugin on ImageJ.</i>
40	2017	Coherency map	materials/nanofibers	Chou et al. (2017)	Journal of the Mechanical Behavior of Biomedical Materials	10.1016/j.jmbbm.2016.09.004	<i>Image analysis was carried out by image processing software (ImageJ, National Institutes of Health, Bethesda, MD, USA) using OrientationJ plugin for fiber alignment measurement by applying a Riesz filter at 70% minimal coherency and 2% minimal energy.</i>
41	2017	Distribution (directionality histogram)	muscle myoblast differentiation	Gatta et al. (2017)	Oncotarget	10.18632/oncotarget.23044	<i>A segment crossing the longest part of each cell was added and used to analyze the distribution of cell direction by the OrientationJ (Distribution) plug-in of Image J.</i>
42	2017	Distribution (directionality histogram)	muscle tissue engineering	Costantini et al. (2017)	Frontiers in Bioengineering and Biotechnology	10.3389/fbioe.2017.00022	<i>The distributions of myotube orientation within the hydrogel structures following in vitro culture was obtained by analyzing IF micrographs (MHC signal) with OrientationJ plugin</i>
43	2017	orientation/coherency analysis	nerve guidance conduit / neural tissue engineering	Peng et al. (2017)	International Journal of Nanomedicine	10.2147/ijn.s122017	<i>used ImageJ and OrientationJ software to estimate cell length and growth direction, respectively</i>
44	2017	Color-coded orientation map	neuronal alignment in hippocampal tissue engineering	Kim et al. (2017)	Nature Communications	10.1038/ncomms14346	<i>OrientationJ plug-in of ImageJ software was used for quantitative analyses of confocal fluorescence images.</i>
45	2017	Distribution (directionality histogram)	neuronal alignment in hydrogels	Rose et al. (2017)	Nano Letters	10.1021/acs.nanolett.7b01123	<i>The orientation time was determined via a software-based orientation analysis (ImageJ plugin OrientationJ), which evaluated every pixel of the image based on a structure tensor</i>
46	2017	Coherency measure	white matter microstructure, injured brain	van et al. (2017)	Scientific Reports	10.1038/s41598-017-16797-1	<i>Coherency of myelinated axons, as an inverse measure of complexity of white matter microstructure, was assessed using the 'OrientationJ measure' function</i>
47	2018	Color-coded orientation map	actin cytoskeleton	Kelley et al. (2018)	Molecular Biology of the Cell	10.1091/mbc.e18-01-0056	<i>OrientationJ (Rezakhaniha et al. , 2012), configured using a Gaussian fit with a pixel size of 2, was used to measure the orientation distribution of actin bundles in individual cells and to generate color-coded images where color indicates orientation, hue indicates coherency, and brightness is the brightness of...</i>
48	2018	Coherency map	actin cytoskeleton	Cascione et al. (2018)	International Journal of Molecular Sciences	10.3390/ijms20010108	<i>the actin fiber coherency parameter was quantified using the OrientationJ plugin</i>
49	2018	Color-coded orientation map	actin cytoskeleton	Wirshing et al. (2018)	Molecular Biology of the Cell	10.1091/mbc.e18-06-0347	<i>OrientationJ (Rezakhaniha et al. , 2012), configured using a Gaussian fit with a pixel size of 2, was used to measure the orientation distribution of actin bundles in individual cells and to generate color-coded images where color indicates orientation, hue indicates coherency, and brightness is the brightness of...</i>
50	2018	Color-coded orientation map	actin cytoskeleton, muscle myofibrils	Pandey et al. (2018)	Developmental Cell	10.1016/j.devcel.2017.12.024	<i>Actin and α-actinin images of single cells were then analysed for filament alignment using the OrientationJ plugin for ImageJ</i>
51	2018	Distribution (directionality histogram)	bone/mineralized tissue	Petersen et al. (2018)	Nature Communications	10.1038/s41467-018-06504-7	<i>Distribution of fiber orientation was analyzed via the ImageJ plug-in OrientationJ (by Daniel Sage) on two to three technical replicates (scaffolds) and three confocal image stacks per scaffold.</i>
52	2018	Dominant Direction	cardiac tissue engineering	Hansen et al. (2018)	Frontiers in Cardiovascular Medicine	10.3389/fcvm.2018.00052	<i>Alpha-Actinin Fiber Alignment Confocal images of alpha-actinin stained hPS-CM seed on microthreads and TCP were uploaded to ImageJ (NIH) for actinin fiber alignment using OrientationJ.</i>
53	2018	structure tensor	cellular nematics, liquid crystals	Duclos et al. (2018)	Nature Physics	10.1038/s41567-018-0099-7	<i>The orientation field was obtained by computing the local structure tensor with ImageJ plugin OrientationJ</i>
54	2018	Coherency map	collagen architecture in scar tissue	Clemons et al. (2018)	RSC Advances	10.1039/c7ra12693j	<i>the OrientationJ plugin gives a value for coherency without the disadvantages previously discussed involving the Gaussian fit</i>
55	2018	Dominant Direction	collagen scaffold tissue engineering	Ashworth et al. (2018)	Journal of Materials Science: Materials in Medicine	10.1007/s10856-018-6175-9	<i>The OrientationJ plugin in FIJI was used to quantify the collagen pore wall orientation from bright field images taken of the scaffold cross-sections.</i>
56	2018	Coherency map	collagen/ECM in tendon tissue	Kinugasa et al. (2018)	Scientific Reports	10.1038/s41598-018-22661-7	<i>The local collagen fiber alignment was determined with the ImageJ plug-in OrientationJ (Version 19.11.2012., available online at http://bigwww.epfl.ch/demo/orientation/)</i>
57	2018	orientation/coherency analysis	collagen/ECM in tumors	Zanotelli et al. (2018)	Molecular Biology of the Cell	10.1091/mbc.e17-01-0041	<i>Fiber orientation was then analyzed in ImageJ using OrientationJ.</i>
58	2018	Orientation angle map	endothelial cell cytoskeleton	Martin et al. (2018)	eLife	10.7554/elife.33864	<i>local orientation angle map was calculated for each pixel using OrientationJ plugin (Püspöki et al., 2016).</i>
59	2018	Dominant Direction	endothelial cell cytoskeleton under shear stress	Morley et al. (2018)	MHR: Basic science of reproductive medicine	10.1093/molehr/gay033	<i>OrientationJ plugin³⁷ quantified cell orientation relative to the direction of shear stress</i>
60	2018	Dominant Direction	endothelial cell network tissue engineering	Skrzypek et al. (2018)	Journal of Materials Science: Materials in Medicine	10.1007/s10856-018-6102-0	<i>The orientation of skeletonized HUVEC structures relative to surface topography (x-axis) was quantified using the OrientationJ plugin.</i>
61	2018	Collagen orientation quantification	intervertebral disc collagen/ECM	Krishnamoorthy et al. (2018)	Disease Models & Mechanisms	10.1242/dmm.036012	<i>Collagen orientation was quantified from the B-SHG images using OrientationJ, a series of plugins for ImageJ</i>
62	2018	structure tensor	materials/nanofibers	Rychter et al. (2018)	Pharmaceutical Research	10.1007/s11095-017-2314-0	<i>Visual orientation analysis was conducted by OrientationJ and involved the structure tensor calculations for each pixel in the SEM image.</i>
63	2018	Distribution (directionality histogram)	muscle mitochondrial network organization	Bleck et al. (2018)	Nature Communications	10.1038/s41467-018-07676-y	<i>the OrientationJ Distributions plugin was used to determine the angles of the mitochondrial network</i>
64	2018	Coherency map	neuronal alignment in white matter	van et al. (2018)	Glia	10.1002/glia.23216	<i>Coherency of NF200 + axons was measured using the OrientationJ plugin for ImageJ (Fonck et al., 2009).</i>
65	2018	structure tensor algorithm	neuronal fiber alignment in brain	Morawski et al. (2018)	NeuroImage	10.1016/j.neuroimage.2017.11.061	<i>the in-plane orientation of fiber structures was obtained from the probability maps using the local edge orientation computed by the Directionality module in FIJI or alternatively using the estimation of local structure tensor algorithm implemented in OrientationJ</i>
66	2018	Coherency map	smooth muscle tissue engineering	Kobayashi et al. (2018)	Scientific Reports	10.1038/s41598-018-31992-4	<i>the plugin OrientationJ was used to conduct coherency analysis (Kit, MHC, GFP and β-tubulin)</i>

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67	2018	Coherency map	vessel/vascular tissue	Gurevich et al. (2018)	The EMBO Journal	10.15252/emboj.201797786	Vessel orientation was measured using the ImageJ plugin OrientationJ, which rates the coherence of neighbouring pixels between 0 (isotropic, no overall direction) and 1 (one dominant direction; Fonck et al , 2009).
68	2019	Distribution (directionality histogram)	actin cytoskeleton	Colin-York et al. (2019)	Communications Biology	10.1038/s42003-019-0322-9	Using the 'OrientationJ' Java plugin for ImageJ, the distribution of fiber orientations within a region of interested was evaluated
69	2019	orientation analysis	actin cytoskeleton	Fearing et al. (2019)	The FASEB Journal	10.1096/fj.201802725rrr	F-actin orientation was analyzed from microscopic images (ImageJ plugin, OrientationJ).
70	2019	Coherency	actin cytoskeleton	Cascione et al. (2019)	International Journal of Molecular Sciences	10.3390/ijms20143594	The cytoskeletal actin organization was characterized in terms of coherency by means of OrientationJ plugin (ImageJ software)
71	2019	orientation/coherency analysis	artificial cells, tubulin and collagen	Fanalista et al. (2019)	ACS Nano	10.1021/acsnano.9b00220	the values of tubulin and collagen bundle orientations were obtained using Fiji (OrientationJ plugin) after the background was appropriately subtracted
72	2019	Vector Field	bacterial biofilm active matter	Yaman et al. (2019)	Nature Communications	10.1038/s41467-019-10311-z	The ImageJ plugin OrientationJ was used to find the nematic director field of the colony.
73	2019	orientation/coherency analysis	biomaterials fiber analysis	Shkarin et al. (2019)	PLOS ONE	10.1371/journal.pone.0215137	The OrientationJ plug-in is aimed initially at the analysis of microscopy images and provides information about orientation, coherency, and energy within a specified ROI.
74	2019	Color-coded orientation map	cell morphology and adhesion	Gonzalez-Molina et al. (2019)	Scientific Reports	10.1038/s41598-019-44960-3	Pseudo-coloured images were obtained with the OrientationJ plugin of ImageJ using a Gaussian window of 2 pixels, a Fourier gradient, and colours representing orientation (hue) and coherency (saturation).
75	2019	Coherency map	cell morphomechanics, nanoparticle effects	De et al. (2019)	Nanoscale Research Letters	10.1186/s11671-019-2941-y	OrientationJ plugin was used to quantify the coherency parameter by choosing a specific sequence of ROIs in confocal acquisitions, based on the measure of the structure tensors in a local neighborhood.
76	2019	Color-coded orientation map, Coherency map, Distribution (directionality histogram)	collagen/ECM biomaterial engineering	Taufalele et al. (2019)	PLOS ONE	10.1371/journal.pone.0216537	To measure the alignment of the collagen matrices, confocal reflectance images were analyzed via the OrientationJ plugin in ImageJ
77	2019	Measure	collagen/ECM in joint osteoarthritis	Reed et al. (2019)	PLOS ONE	10.1371/journal.pone.0214072	All data was collected using the OrientationJ plugin [52] and collagen orientation measurements follow published methods [53].
78	2019	orientation/coherency analysis	collagen/ECM in mammary tissue	Jones et al. (2019)	Neoplasia	10.1016/j.neo.2018.10.010	The maximum intensity projection of these images was analyzed with the OrientationJ plugin for ImageJ [26] , which analyzes fiber orientation on a pixel-by-pixel basis
79	2019	Coherency map	collagen/ECM in tumors	Desa et al. (2019)	Journal of Biomedical Optics	10.1117/1.jbo.24.8.086503	Coherency-based F/B was calculated from tumor bulk and tumor–host interface in each HER2+ core needle biopsy (n = 29) using OrientationJ.
80	2019	orientation/coherency analysis	collagen/ECM in tumors	Natal et al. (2019)	Scientific Reports	10.1038/s41598-019-44156-9	Quantitative analysis of collagen parameters was performed in SHG images using ImageJ and OrientationJ plug-in
81	2019	orientation/coherency analysis	collagen/ECM in tumors	Nguyen et al. (2019)	Molecular & Cellular Proteomics	10.1074/mcp.ra119.001496	The orientation of ECM fibers was analyzed using ImageJ (NIH) plugin OrientationJ as previously described
82	2019	Coherency	collagen/ECM in tumors	Sheen et al. (2019)	eLife	10.7554/elife.45120	coherency, the degree to which the local features are oriented, was measured with OrientationJ (Rezakhaniha et al., 2012)
83	2019	orientation/coherency analysis	collagen/ECM in tumors	Zeltz et al. (2019)	Cancers	10.3390/cancers11050705	Alignment of collagen fibers in the remodeled matrix was recorded using second harmonic generation and analyzed with OrientationJ.
84	2019	orientation/coherency analysis	colloidal liquid crystals, tactoids	Gârlea et al. (2019)	Scientific Reports	10.1038/s41598-019-56729-9	The local alignment is determined using the plug-in "OrientationJ" of "ImageJ".
85	2019	Harris corner detection (with OrientationJ for image analysis support)	fish olfactory organ morphology	Ferrando et al. (2019)	Frontiers in Physiology	10.3389/fphys.2019.00245	this count was performed individuating the Harris corners (by an algorithm) in the lamellae silhouettes using the OrientationJ plug-in for ImageJ
86	2019	average fiber diameter and porosity analysis	materials/nanofibers	Liverani et al. (2019)	Scientific Reports	10.1038/s41598-018-37640-1	The calculation of the average fiber diameter and the analysis of the porosity were performed by using ImageJ and its plugins (DiameterJ and OrientationJ)
87	2019	Orientation/coherency analysis	muscle tissue engineering, meat analogs	MacQueen et al. (2019)	npj Science of Food	10.1038/s41538-019-0054-8	Analysis of fiber diameter and alignment ImageJ software (NIH) with the DiameterJ and OrientationJ plug-ins was used to determine fiber diameter and alignment from the SEM images
88	2019	Coherency map; Color-coded orientation map	neuronal alignment in cerebral organoids	Giandomenico et al. (2019)	Nature Neuroscience	10.1038/s41593-019-0350-2	The product of OrientationJ coherency and energy output levels was quantified in whole organoids compared with ALI-COs, demonstrating more aligned structures.
89	2019	orientation	septin filaments, membrane structure	Beber et al. (2019)	Nature Communications	10.1038/s41467-019-08344-5	SEM images were analyzed using OrientationJ (a plug-in for imageJ) to determine the orientation of the filaments
90	2020	Coherency map	actin cytoskeleton	Layton et al. (2020)	Biology Open	10.1242/bio.049809	The orientation properties of the actin filaments in myofibroblast immunofluorescence images (Phalloidin-488) were computed based on the evaluation of the structure tensor in a local neighbourhood using the Java plugin for ImageJ 'OrientationJ'.
91	2020	Coherency map	actin cytoskeleton	Rougerie et al. (2020)	Scientific Reports	10.1038/s41598-020-70907-0	We quantified the average spatial orientation of actin fibers in confocal z-stack using OrientationJ Fiji plugin 25 and we observed that the anisotropic density of F-actin was accompanied with a curvature-dependent longitudinal alignment
92	2020	orientation/coherency analysis	actin cytoskeleton	Xu et al. (2020)	Frontiers in Cell and Developmental Biology	10.3389/fcell.2020.547342	The anisotropy of actin fibers was measured using OrientationJ plug-in for ImageJ to evaluate the arrangement of stress fibers.
93	2020	Dominant Direction	actin cytoskeleton, skeletal muscle	Owens et al. (2020)	International Journal of Molecular Sciences	10.3390/ijms22010306	Actin fiber coherence in the dominant direction as determined by analysis of confocal images of myotubes stained fluorescently for actin (phalloidin) in ImageJ using OrientationJ plug-in.
94	2020	Distribution (directionality histogram)	arterial tissue mechanics	Weiss et al. (2020)	Science Advances	10.1126/sciadv.abd3574	the in-plane distribution of orientations was estimated using a 2D structure tensor analysis with OrientationJ implemented for ImageJ.
95	2020	structure tensor analysis	astrocyte and myelin alignment	Sullivan et al. (2020)	Acta Neuropathologica Communications	10.1186/s40478-020-00960-3	OrientationJ Plug-in (RRID:SCR_014796, http://bigwww.epfl.ch/demo/orientation/) was used for structure tensor analysis
96	2020	Distribution (directionality histogram)	bone/mineralized tissue	Lim et al. (2020)	Polymers	10.3390/polym13010060	aligned scaffolds showed a distribution of orientations concentrated over 1000 in certain degree orientations, while random patterned scaffolds showed a distribution of orientations under 1000 with varying degree orientations
97	2020	Distribution (directionality histogram)	cardiac myofibril alignment	Chanthra et al. (2020)	Scientific Reports	10.1038/s41598-020-61163-3	immunofluorescent images for α-actinin were transformed into frequency spectrum images for the evaluation of the distributions of orientation frequencies by OrientationJ
98	2020	orientation/coherency analysis	cardiac tissue engineering	Tsukamoto et al. (2020)	Scientific Reports	10.1038/s41598-020-59371-y	ImageJ plug-in, OrientationJ, was used for the quantification of the fabricated 3D tissue's cell orientation
99	2020	Coherency	cardiac tissue engineering	Zouhair et al. (2020)	Biomolecules	10.3390/biom10030371	Coherency, i.e., C, was calculated for collagen and elastin to verify the local dominant orientation in representative regions of interest (n = 9) using OrientationJ

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100	2020	orientation/coherency analysis	cardiac tissue, pulmonary hypertension	Sharifi et al. (2020)	Journal of the American Heart Association	10.1161/jaha.119.015708	<i>Image processing and fiber orientation analysis were performed using the OrientationJ toolbox in ImageJ</i>
101	2020	orientation/coherency analysis	cellular contractile forces and cytoskeleton	Feld et al. (2020)	Science Advances	10.1126/sciadv.aaz6997	<i>Fiber orientation analysis was performed using the Ridge Detection and OrientationJ plugins for Fiji.</i>
102	2020	Color-coded orientation map	cerebral ischemia tissue repair and microvasculature	Lu et al. (2020)	Journal of Biomedical Optics	10.1117/1.jbo.25.4.046002	<i>Local orientation and coherence were computed for each pixel of the filter-enhanced 2-D en face angiogram based on its structure tensor using OrientationJ</i>
103	2020	Coherency map	collagen/ECM in tissue	Liu et al. (2020)	Frontiers in Bioengineering and Biotechnology	10.3389/fbioe.2020.00198	<i>the alignment for OrientationJ uses the coherency calculated in the tool, while the alignment for CytoSpectre equals 1 - circular variance of the orientations.</i>
104	2020	Distribution (directionality histogram)	collagen/ECM in tumors	Hoang et al. (2020)	Frontiers in Oncology	10.3389/fonc.2020.01164	<i>the Fiji plugins DiameterJ and OrientationJ were used to generate data on the distributions of fiber diameter, orientation, and pore size</i>
105	2020	Distribution (directionality histogram)	collagen/ECM in tumors	Jaeschke et al. (2020)	Materials Today Bio	10.1016/j.mtbio.2020.100073	<i>The OrientationJ plugin in Fiji (ImageJ 1.52p) was used for orientation distribution analysis.</i>
106	2020	orientation/coherency analysis	collagen/ECM in tumors	Druzhkova et al. (2020)	International Journal of Molecular Sciences	10.3390/ijms21218119	<i>Quantitative analysis of the SHG signal was performed using ImageJ software (National Institutes of Health, Bethesda, MD, USA) and the OrientationJ plugin.</i>
107	2020	Coherency map	extracellular vesicles in colon cancer	Lucchetti et al. (2020)	Cancers	10.3390/cancers12020260	<i>Analyses were performed using the OrientationJ plugins to analyze the coherency</i>
108	2020	Distribution (directionality histogram)	fibrin ECM and cell mechanotransduction	Natan et al. (2020)	Molecular Biology of the Cell	10.1091/mbc.e20-01-0079	<i>Orientation distribution of the fibrous matrix between cell pairs was conducted based on 2D images, using ImageJ with the ImageJ plug-in, OrientationJ</i>
109	2020	Color-coded orientation map	fibroblast cell alignment and ECM	Hashimoto et al. (2020)	Journal of Biological Physics	10.1007/s10867-020-09556-3	<i>These images were analysed with the ImageJ plugin OrientationJ [19] to generate the colour-coded maps shown in Fig.</i>
110	2020	Measure	lymphatic endothelial cell alignment	Michalaki et al. (2020)	Communications Biology	10.1038/s42003-019-0732-8	<i>0, 6, 12, 18, and 24 h, using the OrientationJ Measure plugin in Fiji for 50 cells per binned region.</i>
111	2020	orientation/coherency analysis	materials/nanofibers	Andrzejewski et al. (2020)	Materials	10.3390/ma13163496	<i>Basalt fiber arrangement was analyzed using the OrientationJ plugin, where the task of this software is to characterize the orientation in the image based on the evaluation of the structure tensor in the local neighborhood.</i>
112	2020	Measure	microtubule motor protein dynamics	Kaneko et al. (2020)	Science Advances	10.1126/sciadv.aax7413	<i>Using OrientationJ, the angles were calculated for each pixel of the fluorescence images of the microtubules.</i>
113	2020	Dominant Direction	plant cell microtubule cytoskeleton	Colin et al. (2020)	Proceedings of the National Academy of Sciences	10.1073/pnas.2008895117	<i>We first calculated a mean CMT orientation per protoplast, with the OrientationJ, Dominant Direction plugin, and second, we calculated a deformation ratio</i>
114	2020	Color-coded orientation map	skeletal muscle myofibrils	Jensen et al. (2020)	Scientific Reports	10.1038/s41598-020-69936-6	<i>Composite images were composed of 252 frames, and the OrientationJ ImageJ plug-in was used to false-color the image according to the local angle of orientation of F-actin filaments.</i>
115	2020	Distribution (directionality histogram)	skeletal muscle tissue engineering	Velasco-Mallorquí et al. (2020)	Nanoscale Advances	10.1039/d0na00268b	<i>The ImageJ PlugIn "OrientationJ" software was used to see the distribution of the aligned fibers and the alignment of the cells in a random fiber distribution of the cryogel against the anisotropic cryogels.</i>
116	2020	Measure	smooth muscle cell alignment, tissue engineering	Adamowicz et al. (2020)	Scientific Reports	10.1038/s41598-020-62197-3	<i>The orientation of cultivated SMC was established by a square ROI (80 × 80 pixels) (n = 10) analysis using ImageJ OrientationJ (EPFL, Switzerland) plugin.</i>
117	2020	Coherency map	tendon tissue engineering, collagen alignment	Talò et al. (2020)	Annals of Biomedical Engineering	10.1007/s10439-019-02257-6	<i>To quantify the coherency and the local alignment of the new collagen deposition within the inner injected channels, ImageJ plug-in OrientationJ (Version 19.11.2012) was employed.</i>
118	2021	Coherency map	actin cytoskeleton	De et al. (2021)	Molecules	10.3390/molecules26102885	<i>Coherency values of F-actin were performed on 20 cells, using the OrientationJ plugin of the ImageJ 1.47 software.</i>
119	2021	Coherency map	actin cytoskeleton	De et al. (2021)	Cancers	10.3390/cancers13143610	<i>Morphometric quantification (coherency of F-actin) was performed on 15 cells using the OrientationJ plugin of the ImageJ 1.47 software.</i>
120	2021	Color-coded orientation map	actin cytoskeleton and mitochondrial networks	Moore et al. (2021)	Nature	10.1038/s41586-021-03309-5	<i>Orientation maps of the segmented cables were generated in the Fiji plugin OrientationJ, using a cubic spline gradient with gaussian window sigma = 2 pix.</i>
121	2021	Distribution (directionality histogram)	actin cytoskeleton in kidney tubules	Myram et al. (2021)	Frontiers in Bioengineering and Biotechnology	10.3389/fbioe.2021.624553	<i>Distribution of F-actin local orientation as assessed by OrientationJ software for PCT Pkd1 +/- (blue) and Pkd1 -/- (red) cells.</i>
122	2021	Color-coded orientation map	actin cytoskeleton, muscle regeneration	Girardi et al. (2021)	Nature Communications	10.1038/s41467-020-20289-8	<i>These pictures were analyzed with OrientationJ (ImageJ Plug-in) to obtain a color-coded orientation mask.</i>
123	2021	Distribution (directionality histogram)	airway smooth muscle tissue engineering	Jin et al. (2021)	Frontiers in Molecular Biosciences	10.3389/fmolb.2021.717771	<i>Assessment of Cell Orientation Quantification of ASMC orientation cultured on the concave and convex surfaces and the cylinder axis was performed using the distribution analysis function in the plugin OrientationJ</i>
124	2021	Color-coded orientation map	cancer cell invasion in microfluidics	Samandari et al. (2021)	Scientific Reports	10.1038/s41598-021-89635-0	<i>the color maps were generated using the "OrientationJ" plugin in ImageJ</i>
125	2021	orientation/coherency analysis	cardiac myocardial fiber kinematics	Sharifi et al. (2021)	Scientific Reports	10.1038/s41598-021-83154-8	<i>Image processing and fiber orientation analysis of histological sections were performed using the OrientationJ toolbox in ImageJ</i>
126	2021	Distribution (directionality histogram)	cardiac tissue, aortic microstructure	Cavinato et al. (2021)	Frontiers in Cardiovascular Medicine	10.3389/fcvm.2021.800730	<i>the in-plane primary orientation and alignment were estimated via a distribution of orientations for each circumferential-axial section</i>
127	2021	Coherency map	cardiac valve fibrosis	Jannasch et al. (2021)	Scientific Reports	10.1038/s41598-021-85142-4	<i>The degree of organization of collagen fibers was retrieved using the plugin OrientationJ for ImageJ.</i>
128	2021	Orientation and Coherency analysis	collagen fiber alignment in ear tissue	Wu et al. (2021)	Frontiers in Cell and Developmental Biology	10.3389/fcell.2021.720383	<i>OrientationJ in ImageJ was used to quantify the orientation and coherency features of the collagen fibers constituting the pars tensa across the thickness</i>
129	2021	Measure	collagen/ECM fibrillogenesis	Jones et al. (2021)	PLOS ONE	10.1371/journal.pone.0245653	<i>fiber orientation was quantified using the ImageJ plugin OrientationJ [25] with previously published settings [24]</i>
130	2021	Color-coded orientation map	collagen/ECM in adipose tissue	Stocco et al. (2021)	Frontiers in Medicine	10.3389/fmed.2021.661403	<i>OrientationJ is a Java plugin for ImageJ/FIJI which allows to visualize with a Hue, Saturation and Brightness (HSB) map the directionality within the sample.</i>
131	2021	Dominant Direction + Coherency	collagen/ECM in tumors	Drain et al. (2021)	Journal of Experimental Medicine	10.1084/jem.20191360	<i>Fiber coherency was measured in the dominant fiber direction as determined by OrientationJ for each image.</i>
132	2021	orientation/coherency analysis	collagen/ECM in tumors	Budden et al. (2021)	Nature Communications	10.1038/s41467-021-22953-z	<i>Fibre orientation analysis was performed using ImageJ OrientationJ plugin</i>
133	2021	Color-coded orientation map	collagen/ECM in wound healing	Parvanian et al. (2021)	International Journal of Molecular Sciences	10.3390/ijms22094678	<i>Color maps of the collagen I fibers generated using the OrientationJ plugin.</i>
134	2021	Distribution (directionality histogram)	endothelial cell cytoskeleton alignment	Silvani et al. (2021)	Frontiers in Bioengineering and Biotechnology	10.3389/fbioe.2021.612151	<i>Distribution of EC orientations after 4 h (green), 24h (red), and 48 h (blue) from seeding obtained with "orientationJ distribution" of ImageJ software.</i>

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135	2021	Color-coded orientation map, Distribution (directionality histogram), Dominant Direction	fibronectin ECM assembly, fibroblast mechanobiology	Garrison et al. (2021)	Molecular Biology of the Cell	10.1091/mbc.e20-08-0533	individual matrix assembly site orientations were calculated for individual cells using OrientationJ, averaged across cells (WI-38 n = 27, NIH 3T3 n = 36), and plotted from −90° to 90°.
136	2021	structure tensor (orientation/coherency analysis)	materials/nanofibers	Tagliaferri et al. (2021)	ACS Nano	10.1021/acsnano.1c06535	The platelet orientation angle was quantified calculating the gradient structure tensor of the tomographic images (via the OrientationJ plug-in for ImageJ)
137	2021	Distribution (directionality histogram) + Color-coded orientation map	materials/nanofibers	Belousov et al. (2021)	Materials	10.3390/ma14216714	The analysis of SEM images in the ImageJ program using the OrientationJ plugin was performed in the Orientation Distribution module with the Gaussian Gradient structural tensor.
138	2021	Distribution (directionality histogram)	microtubule cytoskeleton dynamics	Memarian et al. (2021)	Proceedings of the National Academy of Sciences	10.1073/pnas.2117107118	ImageJ plugin OrientationJ to calculate angular distributions of MTs with a bin size of 5° to calculate angular distributions of the MTs using the published method from Inoue et al.
139	2021	Coherency map / Measure	myelination complexity in neuropsychiatric tissue	Hsieh et al. (2021)	Brain Pathology	10.1111/bpa.12870	Quantitative coherency analysis using OrientationJ to assess the myelination complexity in the ACC and the hippocampal regions.
140	2021	orientation/coherency analysis	neural tissue engineering scaffolds	Osorio-Londoño et al. (2021)	Polymers	10.3390/polym13223876	Cell morphology and orientation of cell structures analysis was conducted in ImageJ/Fiji (NIH, Bethesda, MD, USA), with the OrientationJ plugin
141	2021	orientation/coherency analysis	neuronal alignment, nanofiber scaffolds	Antonova et al. (2021)	Nanomaterials	10.3390/nano11020516	Fiber orientation was determined using the OrientationJ plugin and Directionality ImageJ software
142	2021	Coherency map	neuronal tissue and brain imaging	Alexander et al. (2021)	Theranostics	10.7150/thno.53474	Percent area and coherency were quantified using ImageJ with the OrientationJ plugin as described before
143	2021	Coherency and fiber alignment (Measure)	optic nerve gliosis / neuroinflammation	Tribble et al. (2021)	Translational Vision Science & Technology	10.1167/tvst.10.1.21	GFAP fiber alignment (to detect process remodeling typical of gliosis in the optic nerve) was measured using OrientationJ
144	2022	Coherency map	actin cytoskeleton	Potolitsyna et al. (2022)	Scientific Reports	10.1038/s41598-022-14296-6	Actin cytoskeleton coherence was measured from phalloidin staining with the OrientationJ plugin
145	2022	Coherency map	actin cytoskeleton	Knox et al. (2022)	iScience	10.1016/j.isci.2022.105648	The coherence of the actin filamentous directionality was measured using OrientationJ, a plugin for Fiji/ImageJ, with the region of interest (ROI) set to the entirety of each im...
146	2022	orientation/coherency analysis	actin cytoskeleton	Tsai et al. (2022)	Science Advances	10.1126/sciadv.abp8677	we performed image analysis using the "OrientationJ" plugin in ImageJ/Fiji
147	2022	Color-coded orientation map	airway mucus and mucins	Kato et al. (2022)	Science Advances	10.1126/sciadv.abm9718	Inset was falsely colored according to angular alignment of pixels quantified with OrientationJ.
148	2022	Coherency map	biomaterial scaffold characterization	Casarin et al. (2022)	International Journal of Molecular Sciences	10.3390/ijms23052826	the coherency (C) parameter, to estimate the local orientation of the fibres (values around 0 indicate isotropic areas, whereas values around 1 indicate highly orien...
149	2022	Coherency map	cardiac tissue engineering	Todesco et al. (2022)	Biomimetics	10.3390/biomimetics7030104	Coherency (C) was calculated for collagen and elastin to check the local dominant fiber orientation in the Z-stack images using OrientationJ, an ImageJ plugin
150	2022	Dominant Direction	cardiac tissue microstructure	Kim et al. (2022)	Frontiers in Cardiovascular Medicine	10.3389/fcvm.2022.1037500	Fiber orientation was calculated using the OrientationJ (33) plugin for Fiji at 10 × 10 pixel blocks based on structure tensor analysis.
151	2022	Coherency map	cardiovascular tissue engineering, extracellular matrix	Palmosi et al. (2022)	Frontiers in Bioengineering and Biotechnology	10.3389/fbioe.2022.1042434	The coherence of collagen fibers was calculated to verify the dominant local orientation of the images using OrientationJ
152	2022	Measure	cell alignment in hydrogel scaffolds	Tavakoli et al. (2022)	Molecules	10.3390/molecules27031002	OrientationJ plugin was used to measure the orientation of cells in the input binary (8-bit) microscope images.
153	2022	Distribution (directionality histogram), Coherency parameter, Color-coded orientation map	collagen/ECM in ligament tissue	Kim et al. (2022)	Journal of Experimental Orthopaedics	10.1186/s40634-022-00507-6	The output from the OrientationJ analysis, in terms of distribution and dominant direction, was the histogram of orientation angles, a coherency parameter and a color-coded map of fiber orientations.
154	2022	Distribution (directionality histogram)	collagen/ECM in tumors	Alfonso et al. (2022)	Biomedicines	10.3390/biomedicines11010019	Analysis using the OrientationJ plugin in ImageJ indicates a similar distribution of fiber angles between the control and BmpR1a ΔFoxL1+ mice in the middle part of the glands
155	2022	orientation/coherency analysis	collagen/ECM in tumors	Jenkins et al. (2022)	Aging Cell	10.1111/accel.13665	SHG images were then analyzed using OrientationJ in ImageJ
156	2022	Distribution (directionality histogram)	collagen/ECM in tumors	Koorman et al. (2022)	Oncogene	10.1038/s41388-022-02258-1	Collagen alignment (collagen bundle angle variability perpendicular to the organoid) by collectively invading H7 organoids was visualized using confocal reflection microscopy (insets) (C) and quantified using OrientationJ in (D).
157	2022	orientation/coherency analysis	collagen/ECM in tumors	Capparelli et al. (2022)	Nature Communications	10.1038/s41467-022-28801-y	Using FIJI's plugin OrientationJ, a comprehensive orientation analysis was conducted using similar plugin settings for all the images evaluated.
158	2022	Measure	hepatic tissue engineering, microvessel orientation	Bonanini et al. (2022)	Angiogenesis	10.1007/s10456-022-09842-9	Maximum intensity projections from each chip were used to calculate the average microvessel orientation and microvessel area using the 'orientationJ analysis' tool
159	2022	Coherency map	implantable drug delivery platform	Whyte et al. (2022)	Nature Communications	10.1038/s41467-022-32147-w	Using the OrientationJ 2.0.5 plugin, the coherency of each ROI was calculated.
160	2022	Distribution (directionality histogram)	intermediate filament cytoskeleton	Shiota et al. (2022)	Frontiers in Cell and Developmental Biology	10.3389/fcell.2022.750829	The bundle alignments of IFs were measured by the dispersion of the IF angles in whole cells using the OrientationJ plugin
161	2022	Dominant Direction	intervertebral disc fibrous tissue	Lintz et al. (2022)	JOR SPINE	10.1002/jsp2.1191	Fiber orientation in the AF region was determined for wild type and diabetic samples using the OrientationJ plugin.
162	2022	Color-coded orientation map	materials/composite microstructure	Liu et al. (2022)	Nature Communications	10.1038/s41467-022-32792-1	The image is colour-coded based on the microplatelet alignment using OrientationJ.
163	2022	Distribution (directionality histogram)	materials/nanofibers	Orlov et al. (2022)	Advances in Skin & Wound Care	10.1097/01.asw.0000823972.164	OrientationJ" of the ImageJ software suite (version 1.X), which segments the fibers in the digital micrographs and calculates the probability function for their planar orientation
164	2022	orientation/coherency analysis	materials/nanofibers	Kratzer et al. (2022)	Cells	10.3390/cells11233774	fiber orientation was defined using the ImageJ plugin OrientationJ
165	2022	Dominant Direction	meniscus collagen/cartilage ECM	Otani et al. (2022)	Scientific Reports	10.1038/s41598-022-16206-2	The FIJI plug-in OrientationJ was used to evaluate collagen orientation.
166	2022	Distribution (directionality histogram)	muscle mitochondrial network organization	Katti et al. (2022)	Nature Communications	10.1038/s41467-022-34445-9	The images were then binarized and the OrientationJ Distributions plugin was used to determine the angles of the mitochondrial network.

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167	2022	Color-coded orientation map & Vector Field	muscle myofibrils / iPSC-derived muscle	Mao et al. (2022)	eLife	10.7554/elife.76649	<i>Vector field analysis of myofiber bundles was performed using the OrientationJ plugin (Rezakhaniha et al., 2012) in Fiji (Schindelin et al., 2012) combined with custom R scripts.</i>
168	2022	Distribution (directionality histogram) and Measure (coherency)	myelinated fiber alignment in cortex	Brocca et al. (2022)	Cells	10.3390/cells11213485	<i>Distribution function of the OrientationJ plugin was used to build a weighted orientation histogram for each ROI</i>
169	2022	orientation/coherency analysis	soft tissue mechanobiology	Stracuzzi et al. (2022)	Biomechanics and Modeling in Mechanobiology	10.1007/s10237-021-01543-4	<i>such as the 'Directionality' tool or the 'OrientationJ' plugin in ImageJ</i>
170	2022	orientation/coherency analysis	tissue dynamics and cellular flows	Thiagarajan et al. (2022)	iScience	10.1016/j.isci.2022.105053	<i>(2012) https://imagej.net/software/fiji/ OrientationJ plugin, Fiji Püspöki et al.</i>
171	2022	Color-coded orientation map	tissue engineered tendon, collagen fibril alignment	Janvier et al. (2022)	Journal of Tissue Engineering	10.1177/20417314221130486	<i>The ImageJ plug-in (OrientationJ) was used to visualise and quantify fibril orientation parallel to the strain axis (0° = parallel, shown as vertical in these images)</i>
172	2022	Coherency map	tissue engineering, collagen/ECM	Maghin et al. (2022)	npj Regenerative Medicine	10.1038/s41536-022-00222-x	<i>Collagen coherency was calculated from SHG to verify the local dominant orientation of the fibers using OrientationJ, an ImageJ plugin</i>
173	2022	Coherency map	white matter axonal alignment	Fu et al. (2022)	Acta Neuropathologica	10.1007/s00401-022-02488-3	<i>The anisotropic properties of axons were characterised by their directional coherency coefficient, which was calculated by the 'OrientationJ' Java plugin for the ImageJ/Fiji software</i>
174	2023	Coherency map, Orientation map, Measure	Vibrio campbellii biofilm morphology	Cesaria et al. (2023)	International Journal of Molecular Sciences	10.3390/ijms24065423	<i>the measured images were processed with OrientationJ to obtain orientation and coherency maps over the whole image extension.</i>
175	2023	Distribution (directionality histogram)	actin cytoskeleton	Gudde et al. (2023)	ACS Applied Bio Materials	10.1021/acsbm.3c00433	<i>distribution of the actin filament orientation on day 21 (ImageJ version 1.53c, plugin OrientationJ)</i>
176	2023	Orientation/coherency analysis	actin cytoskeleton	Liu et al. (2023)	Nature Communications	10.1038/s41467-023-42341-z	<i>we employed the OrientationJ plugin within the Fiji software to determine the orientation of actin filaments.</i>
177	2023	Measure	actin cytoskeleton	Tee et al. (2023)	Nature Communications	10.1038/s41467-023-35918-1	<i>the angle of stress fibre tilt of each elliptical cell, single or in a time-lapse series, was measured using the OrientationJ Measure plugin</i>
178	2023	Orientation map	actin cytoskeleton	Castiglioni et al. (2023)	Nanomaterials	10.3390/nano13050801	<i>Images derived from fluorescence microscopy for F-actin on bMSCs were analyzed using the OrientationJ plugin of ImageJ, as described in detail in [26 , 27], in order to compute an orientation map.</i>
179	2023	Color-coded orientation map	actin cytoskeleton, fibroblast mechanobiology	Angelini et al. (2023)	iScience	10.1016/j.isci.2023.107283	<i>Staining of F-actin shown in false color using the OrientationJ plugin (ImageJ algorithm) depicting the changes in cell polarization and actin fiber orientation</i>
180	2023	Directionality analysis	cardiac muscle myofibrils	Krumbein et al. (2023)	Communications Biology	10.1038/s42003-023-05547-x	<i>For the directionality of pixels in MF20 cardiomyocyte staining, ImageJ software was used, using the OrientationJ plugin</i>
181	2023	Coherency map	cardiac tissue engineering	Casarin et al. (2023)	Journal of Functional Biomaterials	10.3390/jfb14030141	<i>collagen fibre distribution was evaluated through fast Fourier transform (FFT) and coherency (C) parameters, to determine the fibres' local orientation, using the plug-in OrientationJ</i>
182	2023	Directionality (cell orientation analysis)	cardiac tissue engineering	Korover et al. (2023)	Biological Research	10.1186/s40659-023-00442-5	<i>the immunostained iPSCs derived CMs projections were transformed into binary images, and the transformed image was analyzed using OrientationJ for cell directionality.</i>
183	2023	Color-coded orientation map	cartilage tissue engineering	Wang et al. (2023)	Materials Today Bio	10.1016/j.mtbio.2023.100624	<i>OrientationJ plugin from Image J was used to create the color maps of the fiber orientation, while directionality plugin was used to quantify the mean orientation and angular dispersion of the collagen fibers</i>
184	2023	Coherency map	cartilage/meniscal tissue engineering	Barceló et al. (2023)	ACS Biomaterials Science & Engineering	10.1021/acsbiomaterials.3c00107	<i>Analysis on the fiber orientation and coherency was carried out using the OrientationJ plugin in ImageJ software.</i>
185	2023	Coherency map, Orientation angle distribution, Color-coded orientation map	collagen/ECM in bladder carcinoma	Chang et al. (2023)	Journal of Biomedical Optics	10.1117/1.jbo.28.10.102907	<i>The local orientations of collagen fibers were quantified using OrientationJ (Version 2.0.5) with a 6-pixel (~ 1.8 μ m) Gaussian window structure tensor and color-coded on top of the SHG image</i>
186	2023	Coherency map, Dominant Direction	collagen/ECM in pulmonary fibrosis	Hewitt et al. (2023)	Nature Communications	10.1038/s41467-023-41621-y	<i>OrientationJ was used to measure fibre coherency utilising properties of the original image, so-called 'first-order statistics'.</i>
187	2023	orientation/coherency analysis	collagen/ECM in tissues	Kasahara et al. (2023)	Science Advances	10.1126/sciadv.adf9917	<i>The orientation of collagen fibers was characterized by processing confocal images of collagen with OrientationJ ImageJ plugins</i>
188	2023	Distribution (directionality histogram)	collagen/ECM in tumors	He et al. (2023)	Proceedings of the National Academy of Sciences	10.1073/pnas.2209260120	<i>The plugin OrientationJ-Distribution was applied to analyze the fiber orientation distribution automatically.</i>
189	2023	Coherency map; Dominant Direction	corneal collagen fibril organization	Raoux et al. (2023)	Light: Science & Applications	10.1038/s41377-023-01224-0	<i>We use the ImageJ plug-in OrientationJ, which is based on the local evaluation of the gradient structure tensor</i>
190	2023	Dominant Direction	fibrin networks, biomaterial structure	Hedayati et al. (2023)	Chemical Communications	10.1039/d3cc05281h	<i>We used OrientationJ to estimate the local predominant orientation (see ESI †).</i>
191	2023	Vector Field	hair follicle orientation, skin	Cetera et al. (2023)	Development	10.1242/dev.202078	<i>To determine the orientation of each hair follicle, the plug-in 'OrientationJ-Vector Field' was used</i>
192	2023	orientation/coherency analysis	materials/nanofibers	Basar et al. (2023)	ACS Omega	10.1021/acsomega.2c05838	<i>Fiber orientation was analyzed using the OrientationJ plugin to the ImageJ (NIH) software.</i>
193	2023	orientation/coherency analysis	materials/nanofibers	Park et al. (2023)	Scientific Reports	10.1038/s41598-023-29812-5	<i>OrientationJ, which is an ImageJ plug-in, was used to statistically analyze the orientation of PVDF nanofibers.</i>
194	2023	Distribution (directionality histogram)	materials/nanofibers	Taborda et al. (2023)	ACS Omega	10.1021/acsomega.3c06436	<i>the SEM images were analyzed with ImageJ Software (NIH, version 1.50i, Bethesda, MD, USA) with the DiameterJ and OrientationJ plugins</i>
195	2023	orientation/coherency analysis	materials/nanofibers	Cai et al. (2023)	BMC Molecular and Cell Biology	10.1186/s12860-023-00478-1	<i>Fiber orientation was analyzed using the OrientationJ plugin (ImageJ) on 5 different images of the scaffolds taken in 10,000x magnification.</i>
196	2023	Dominant Direction	materials/nanofibers	Lasenko et al. (2023)	Polymers	10.3390/polym15030673	<i>The fiber orientation was obtained using the ImageJ software's OrientationJ plug-in [43 , 44] (ImageJ, National Institutes of Health, Bethesda, MD, USA).</i>
197	2023	Measure	microtubule dynamics	Maan et al. (2023)	Nature Cell Biology	10.1038/s41556-022-01037-0	<i>Each data point represents the average speed on one microtubule estimated using the imageJ plugin orientationJ.</i>
198	2023	Distribution (directionality histogram)	muscle architecture in ultrasound	Bao et al. (2023)	Wearable Technologies	10.1017/wtc.2022.30	<i>The DistributionJ within the OrientationJ plug-in was then utilized to create a table with the distribution of the orientation angle of the targeted aponeurosis and muscle fascicle.</i>
199	2023	Distribution (directionality histogram), Color-coded orientation map, Horizontal Alignment	nanofiber scaffold alignment, cardiovascular tissue engineering	Peters et al. (2023)	APL Bioengineering	10.1063/5.0172423	<i>Each image was processed in ImageJ using OrientationJ's Distribution analysis function which outputs a distribution of angular occurrences from −90° to 90°.</i>

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200	2023	Coherency	nerve conduit fibers, peripheral nerve regeneration	Stocco et al. (2023)	Materials Today Bio	10.1016/j.mtbio.2023.100761	organization/distribution of the fibers was also investigated determining the Coherency (C) by FFT [38] and OrientationJ [39]
201	2023	Color-coded orientation map	septin cytoskeleton organization	Nakazawa et al. (2023)	Journal of Cell Science	10.1242/jcs.260813	The ImageJ plugin OrientationJ displayed the orientation of the filaments on probability maps by color, and a spectrum displayed the angle distribution.
202	2023	Measure	tendon tissue engineering	Rinoldi et al. (2023)	ACS Materials Au	10.1021/acsmaterialsau.3c00012	Cell orientation was quantified based on confocal images using ImageJ (National Institute of Health, USA) by measuring cell cytoskeletons orientation (ImageJ software, OrientationJ).
203	2023	Coherency	tracheal tissue engineering, decellularized grafts	Stocco et al. (2023)	Cells	10.3390/cells12060888	Coherency (C) was determined to assess the local dominant orientation of collagen fibers; to achieve that, it was used the ImageJ plugin OrientationJ
204	2023	orientation/coherency analysis	white matter tissue	Holroyd et al. (2023)	Biomedicines	10.3390/biomedicines11030909	Similar methods were used to quantify white matter orientation by Walsh et al. [2], in this case employing the OrientationJ plugin of ImageJ.
205	2024	Vector Field	Drosophila embryo cell alignment	Ray et al. (2024)	Development	10.1242/dev.202577	Vector fields were obtained with the OrientationJ plugin (local window of 4 pixels for DE-cad, local window of 6 pixels for MTs, and grid size of 10 vectors
206	2024	Coherency map	actin cytoskeleton	Maxey et al. (2024)	APL Bioengineering	10.1063/5.0233737	Cytoskeletal alignment was calculated using actin signal and ImageJ OrientationJ.
207	2024	orientation/coherency analysis	actin cytoskeleton	Jühlen et al. (2024)	Scientific Reports	10.1038/s41598-023-50615-1	actin and focal adhesion orientation was investigated in ImageJ using plugins Analyze Skeleton (2D/3D; version 3.3.0) and OrientationJ Analysis (version 2.0.4), respectively
208	2024	structure-tensor order parameter	actin cytoskeleton	Sakamoto et al. (2024)	Communications Biology	10.1038/s42003-024-06528-4	We measure the orientation field of F-actin using a public domain program OrientationJ implemented as a Fiji/ImageJ plugin.
209	2024	orientation field measurement / Coherency map	actin cytoskeleton	Sakamoto et al. (2024)	Nature Communications	10.1038/s41467-024-53228-y	We measure the orientation field of F-actin using a public domain program OrientationJ implemented as a Fiji/ImageJ plugin.
210	2024	Coherency map	actin cytoskeleton in cancer cells	Popęda et al. (2024)	Experimental & Molecular Medicine	10.1038/s12276-024-01308-w	F-actin coherency was calculated from the projected confocal images using the Orientation J plugin (http://bigwww.epfl.ch/demo/orientationj/) for ImageJ 32 , with the F-actin fiber coherency parameter calculated for selected cells.
211	2024	Color-coded orientation map	biomimetic scaffolds / tissue engineering	Ma et al. (2024)	International Journal of Nanomedicine	10.2147/ijn.s449952	The OrientationJ plug-in for ImageJ was employed to color-code fibers based on their orientation, as shown in Figure 2I and J.
212	2024	Color-coded orientation map	bone and biological fibrous tissues	Muñoz et al. (2024)	Frontiers in Bioengineering and Biotechnology	10.3389/fbioe.2024.1497837	OrientationJ computes the orientations based on the gradient structure tensor in a local neighborhood (Püspöki et al., 2016).
213	2024	orientation/coherency analysis	bone/mineralized tissue cryogels	Castanheira et al. (2024)	ACS Applied Materials & Interfaces	10.1021/acsaami.3c18290	The analysis of the pore orientation was performed resorting to OrientationJ plugin from ImageJ.
214	2024	structure-tensor order parameter	cardiac tissue engineering	Prondzynski et al. (2024)	Nature Communications	10.1038/s41467-024-50224-0	utilizes ImageJ Plugins (OrientationJ) and a custom-made Matlab script for structural analysis of the cells Resulting parameters for sarcomere alignment referred to as "Orientational Order Parameter" (OOP)
215	2024	orientation/coherency analysis	collagen fiber alignment in tendons	Fang et al. (2024)	Science Advances	10.1126/sciadv.ado2015	The disrupted fiber orientation in SHG images was quantitatively analyzed using the "OrientationJ" plug-in of the ImageJ software
216	2024	Orientation analysis	collagen/ECM in Dupuytren's disease	Lenzi et al. (2024)	International Journal of Molecular Sciences	10.3390/ijms25137192	The orientation analysis of collagen structure was performed using the second harmonic generation of collagen fibers images (Z-series of confocal stacks) from control and DD patients built by OrientationJ plugin
217	2024	Color-coded orientation map & Distribution (directionality histogram)	collagen/ECM in tumors	Plesselova et al. (2024)	Cellular and Molecular Bioengineering	10.1007/s12195-024-00817-y	the orientation of the collagen fibers was determined in ImageJ (FIJI) using the OrientationJ plugin for Hue, Saturation and Brightness (HSB) color-survey maps and OrientationJ distribution measurement.
218	2024	Coherency map	collagen/ECM in tumors	Khalil et al. (2024)	Nature Communications	10.1038/s41467-024-49230-z	OrientationJ determines the local orientation (from −90° up to 90°) and coherency (from 0 to 1) of every pixel relative to its neighbor in rectangular regions (30 × 50 μm) of the Collagen I
219	2024	orientation/coherency analysis	collagen/ECM in wound healing	Roy et al. (2024)	Aging Cell	10.1111/acerl.14032	The fiber alignment of collagen I deposited by the implanted cells was computed by the OrientationJ Fiji plugins
220	2024	structure tensor (orientation & coherency)	confluent cell layers	Nejad et al. (2024)	Nature Communications	10.1038/s41467-024-47702-w	Cell orientations were determined using the ImageJ plugin OrientationJ.
221	2024	Color-coded orientation map	dental epithelial tissue morphogenesis	Du et al. (2024)	PLOS Genetics	10.1371/journal.pgen.1011326	Analysis of nuclear orientations in 4 representative E13.5 control incisor germs using OrientationJ reveals cell alignment around the forming enamel knot
222	2024	Coherency	electrospun nanofiber scaffolds	Forouharshad et al. (2024)	RSC Advances	10.1039/d4ra02309a	fiber alignment was quantified as a coherency value by OrientationJ
223	2024	Measure	electrospun nanofibers, tissue engineering	Tai et al. (2024)	Bioengineering	10.3390/bioengineering1108077	DiameterJ and OrientationJ plugins were employed to measure the diameter and orientation of the fibers.
224	2024	Distribution (directionality histogram)	fibrous biomaterial/scaffold orientation	Kim et al. (2024)	Materials Today Bio	10.1016/j.mtbio.2024.101182	PSF and PTFE direction of orientation was measured using OrientationJ Distribution plugin in the Fiji software.
225	2024	orientation/coherency analysis	gingival tissue engineering	Llopis-Grimalt et al. (2024)	ACS Omega	10.1021/acsomega.4c02253	they were analyzed using the FIJI software Plugin OrientationJ
226	2024	orientation/coherency analysis	materials/battery separators	Seo et al. (2024)	Nature Communications	10.1038/s41467-024-54313-y	The structural information of the PE and HM separators, including the fiber thickness, orientation, and particle size, was gathered from their SEM images using ImageJ and OrientationJ software
227	2024	Distribution (directionality histogram)	materials/ceramic composites	Hackert et al. (2024)	Materials	10.3390/ma17102175	The fiber orientation was analyzed with imageJ [30] using the orientationJ plugin with its analysis and distribution functions.
228	2024	Measure	materials/nanofibers	Sousa et al. (2024)	Nanomaterials	10.3390/nano14040342	ImageJ (National Institutes of Health, Bethesda, MD, USA) and the OrientationJ plugin were used to determine the average diameter and orientation of the fibers.
229	2024	Coherency	materials/nanofibers	Pawar et al. (2024)	Manufacturing Letters	10.1016/j.mfglet.2024.09.043	The OrientationJ plugin of ImageJ was used to determine the average fiber alignment, expressed as the degree of angular coherency (%), for each group using the 200X images
230	2024	orientation/coherency analysis	muscle myofibrils	Gao et al. (2024)	Materials Today Bio	10.1016/j.mtbio.2024.101195	we assessed myoblast alignment, a morphological indicator closely associated with the initiation of myotube formation, by quantifying stress fiber orientation using the OrientationJ plugin
231	2024	Vector Field	muscle tissue engineering, myotube-ECM alignment	Skillin et al. (2024)	Science Advances	10.1126/sciadv.adn0235	Myotube alignment and nematic correlation length (ξ nn) were quantified using OrientationJ and custom scripts
232	2024	Color-coded orientation map	neuronal alignment in Alzheimer's pathology	Comrie et al. (2024)	Imaging Neuroscience	10.1162/imag_a_00164	OrientationJ was applied using a structure tensor with a Gaussian window of 2 pixels and a cubic spline gradient algorithm where the hue corresponds to orientation, the color saturation corresponds to the fiber coherency

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233	2024	Color-coded orientation map	neuronal tissue and polarized light imaging	Bonaventura et al. (2024)	Journal of Biomedical Optics	10.1117/1.jbo.29.5.052914	<i>OrientationJ was applied using a structure tensor having a gaussian window of 2 pixels and a cubic spline gradient algorithm where the hue corresponds to orientation, the color saturation corresponds to the fiber coherency</i>
234	2024	Vector Field	plant cell cortical microtubules	Melogno et al. (2024)	Proceedings of the National Academy of Sciences	10.1073/pnas.2320470121	<i>we next analyzed CMT orientation with respect to the stem axis in NPA-treated plants, using the OrientationJ plugin in Fiji.</i>
235	2024	structure tensor	primate brain myelin density	Reveley et al. (2024)	Imaging Neuroscience	10.1162/imag_a_00368	<i>the structure tensor as implemented in the "OrientationJ" ImageJ package using a local Gaussian window $\sigma = 85$ pixels and image gradients computed with a cubic spline</i>
236	2024	Orientation and Coherency	retinal glial cell alignment	Henze et al. (2024)	Frontiers in Neuroscience	10.3389/fnins.2024.1290829	<i>inner microglia in each circle were analyzed for orientation and coherency using the FIJI OrientationJ plug-in relative to the central lesion axis</i>
237	2024	Distribution (directionality histogram)	synthetic cells, peptide-DNA cytoskeleton	Daly et al. (2024)	Nature Chemistry	10.1038/s41557-024-01509-w	<i>Alignment fraction analysis was performed using the OrientationJ Analysis plugin on ImageJ.</i>
238	2025	orientation/coherency analysis	actin cytoskeleton	Elblová et al. (2025)	Journal of Materials Chemistry B	10.1039/d5tb00074b	<i>Cytoskeleton filaments orientation analysis was performed using the ImageJ plugin "OrientationJ".</i>
239	2025	Dominant Direction	actin cytoskeleton	Dessalles et al. (2025)	Nature Physics	10.1038/s41567-025-02847-3	<i>Actin fibre orientation was obtained from the images of phalloidin stainings using the OrientationJ plug-in in ImageJ</i>
240	2025	Vector Field	actin cytoskeleton	Shimolina et al. (2025)	Scientific Reports	10.1038/s41598-025-31162-3	<i>The orientation of actin filaments in cells was analyzed using the structure tensor method via the ImageJ plugin OrientationJ (v2.0.7)</i>
241	2025	Coherency map	actin cytoskeleton	Tarantino et al. (2025)	Discover Nano	10.1186/s11671-025-04186-x	<i>Morphometric quantification (coherency of F-actin) was performed on 15 cells using the OrientationJ plugin of the ImageJ 1.53a software.</i>
242	2025	Coherency map and Measure	actin cytoskeleton and osteoblast adhesion	Valkov et al. (2025)	Journal of Functional Biomaterials	10.3390/jfb17010016	<i>OrientationJ measure was then applied to each ROI to extract coherency values, which served as the primary quantitative metric of cytoskeletal alignment.</i>
243	2025	structure-tensor order parameter	actin cytoskeleton on curved tissue surfaces	Eckert et al. (2025)	Nature Communications	10.1038/s41467-025-62694-x	<i>OrientationJ computes the largest eigendirection of the structure tensor for each pixel, i.e., its intensity orientation, of the image.</i>
244	2025	Distribution (directionality histogram)	bone/mineralized tissue	Hu et al. (2025)	Nature Communications	10.1038/s41467-025-65636-9	<i>dendrite orientation distribution was analyzed with the OrientationJ Plugin in Fiji</i>
245	2025	Coherency map, Color-coded orientation map	cartilage tissue engineering	Jia et al. (2025)	Bioactive Materials	10.1016/j.bioactmat.2025.02.011	<i>OrientationJ, an ImageJ plugin developed in-house based on structure tensors [38].</i>
246	2025	orientation/coherency analysis	cartilage tissue engineering, fibre alignment	Pahoff et al. (2025)	Materials Today Bio	10.1016/j.mtbio.2025.102550	<i>each tomogram slice of the scanned constructs was analysed using OrientationJ in ImageJ software</i>
247	2025	Coherency map	collagen bioresins, deep vat printing	Winkelbauer et al. (2025)	Advanced Healthcare Materials	10.1002/adhm.202405105	<i>Alignment was quantified using the Fiji OrientationJ plugin (available at http://bigwww.epfl.ch/demo/orientationj).</i>
248	2025	Distribution (directionality histogram)	collagen/ECM in tumors	Gao et al. (2025)	Science Advances	10.1126/sciadv.adz4291	<i>Orientation distributions of collagen around intact single microchannels for each cell line (OrientationJ).</i>
249	2025	Coherency map	collagen/ECM in tumors	Sensi et al. (2025)	Translational Oncology	10.1016/j.tranon.2025.102523	<i>Coherency (C) for collagen and elastin was quantified to evaluate the local prevailing orientation of the images, employing OrientationJ.</i>
250	2025	Orientation analysis	collagen/ECM in tumors	Olmedo-Moreno et al. (2025)	Theranostics	10.7150/thno.104143	<i>Orientation analysis of Sirius Red stained samples was performed with the ImageJ plugin OrientationJ.</i>
251	2025	Coherency map; Measure	collagen/actin alignment in epidermis	Oak et al. (2025)	Science Advances	10.1126/sciadv.adq0638	<i>the dominant orientation and coherency of the collagen network were calculated using OrientationJ</i>
252	2025	Distribution (directionality histogram)	dental epithelial-mesenchymal tissue engineering	Huang et al. (2025)	Materials Today Bio	10.1016/j.mtbio.2025.102451	<i>The alignment angle distribution was simulated with the OrientationJ plugin in ImageJ (NIH, Bethesda, MD) normalized by Gaussian nonlinear regression.</i>
253	2025	Distribution (directionality histogram)	extracellular matrix fiber alignment	Thompson et al. (2025)	Journal of Cell Biology	10.1083/jcb.202411167	<i>Alignment is computed by the plug-in OrientationJ, which calculates the proportion of fibers within an image orientated in a similar direction.</i>
254	2025	Dominant Direction	fibre tracing in biomedical images	Arafat et al. (2025)	PLOS ONE	10.1371/journal.pone.0320006	<i>Twombli also incorporates OrientationJ to obtain the global alignment metric, determining the dominant direction of the fibrial structure.</i>
255	2025	Dominant Direction	fibronectin/ECM in glioblastoma	Crestani et al. (2025)	Advanced Science	10.1002/advs.202416526	<i>The function "OrientationJ dominant direction" was then applied to each nucleus with a custom-written macro and their distribution plotted in GraphPad Prism for visualization.</i>
256	2025	Distribution (directionality histogram)	food materials / meat analogue texture	Tang et al. (2025)	Foods	10.3390/foods14142479	<i>SEM images (×1500 magnification) were first converted to 8-bit grayscale and then analyzed using OrientationJ Distribution to extract the angular distribution of fiber orientation.</i>
257	2025	Distribution (directionality histogram)	materials/fiber reinforced concrete	Carrera et al. (2025)	Materials	10.3390/ma19010125	<i>Image analysis was performed with FIJI ImageJ software (version 1.54) and the OrientationJ plugin (version 2.0.7) to objectively quantify orientation angles and distribution histograms.</i>
258	2025	Dominant Direction and Coherency	meat analogues materials	Schlangen et al. (2025)	Current Research in Food Science	10.1016/j.crfs.2025.100980	<i>The dominant direction in each image was determined and the coherency to this dominant direction was calculated with OrientationJ.</i>
259	2025	Distribution (directionality histogram)	muscle tissue engineering	Liu et al. (2025)	Advanced Materials	10.1002/adma.202501147	<i>Then the OrientationJ plugin was used to analyze the orientation distribution of the fibril structures.</i>
260	2025	Color-coded orientation map	muscle tissue engineering	Winkelbauer et al. (2025)	Advanced Science	10.1002/advs.202512727	<i>Confocal images of matured constructs were analyzed in Fiji (ImageJ) using the OrientationJ plugin</i>
261	2025	Vector Field; Color-coded orientation map	muscle tissue engineering; myoblast alignment	Huo et al. (2025)	ACS Nano	10.1021/acsnano.5c08392	<i>The orientation director field and orientation colormap of the cell sheet were obtained using the ImageJ plugin OrientationJ, with both the window size and grid size set to 15 μm</i>
262	2025	orientation/coherency analysis	neuronal alignment and neurite directionality	Hoffmann et al. (2025)	Stem Cell Research & Therapy	10.1186/s13287-025-04591-0	<i>The FIJI OrientationJ plugin developed by Fonck et al. in 2009 was here implemented in a novel region-based configuration to examine neurites emanating from each somata or somata cluster.</i>
263	2025	Distribution (directionality histogram)	neuronal alignment in hydrogels	Liu et al. (2025)	Materials Today Bio	10.1016/j.mtbio.2025.101471	<i>the 'OrientationJ' plugin was used to evaluate the orientation and distribution of the aligned neurites</i>
264	2025	Measure	neuronal axon growth and directionality	De et al. (2025)	Advanced Science	10.1002/advs.202500400	<i>Measures of directionality were obtained using the semi-automated Fiji plugin OrientationJ, with default settings.</i>
265	2025	Distribution (directionality histogram)	neuronal synaptic dendritic alignment	Karunakaran et al. (2025)	Molecular Neurodegeneration	10.1186/s13024-025-00850-z	<i>Maximum intensity projected 60 × images with PSD95 staining were analyzed using the OrientationJ plugin in ImageJ software by selecting the OrientationJ distribution</i>
266	2025	Color-coded orientation map	scaffold materials / tissue engineering	Long et al. (2025)	Advanced Science	10.1002/advs.202417447	<i>OrientationJ" and "Directionality" plugins, from Fiji were used to create the color maps of the pore orientation and to generate data to plot the directionality curves</i>
267	2025	orientation/coherency analysis	vascular tissue, bacterial colonization	Pinon et al. (2025)	eLife	10.7554/elife.107813	<i>For each ROI, the mean fluorescence intensity has been subtracted from the background intensity (collagen matrix) and normalized by the mean of the non-infected areas' intensity</i>

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268	2025	Measure	white matter microstructure, neuronal tissue	Wang et al. (2025)	Pediatric Research	10.1038/s41390-024-03412-z	<i>Coherency of myelinated axons, as an inverse measure of complexity of white matter microstructure, was assessed using the "OrientationJ measure" function of ImageJ plugin OrientationJ</i>
269	2026	Distribution (directionality histogram)	actin cytoskeleton	Jiang et al. (2026)	Biological Research	10.1186/s40659-026-00686-x	<i>For in-depth orientation analysis and quantification of the distribution of phalloidin-stained filaments alignment between both cell types under each experimental condition, we used the software plugin tool OrientationJ (ImageJ/Fiji).</i>
270	2026	structure-tensor order parameter	actin cytoskeleton	Chen et al. (2026)	Cell Death & Disease	10.1038/s41419-026-08678-1	<i>the alignment parameter (S), calculated by OrientationJ software, derived from the gradient structure tensor, was used to assess filaments alignment</i>
271	2026	Coherency map	actin cytoskeleton	Sarmiento-Jiménez et al. (2026)	International Journal of Molecular Sciences	10.3390/ijms27073026	<i>F-actin bundle alignment was quantified using the OrientationJ plugin in Fiji 2.3</i>
272	2026	Distribution (directionality histogram)	actin cytoskeleton	Mondal et al. (2026)	Life Science Alliance	10.26508/lsa.202603664	<i>For each ROI, the filament orientation information was extracted using "OrientationJ (OJ) Distribution (Rezakhaniha et al, 2012)" plugin in Fiji.</i>
273	2026	Color-coded orientation map	actin cytoskeleton	Li et al. (2026)	International Journal of Molecular Sciences	10.3390/ijms27104380	<i>The orientation of the F-actin cytoskeleton was analyzed using the OrientationJ plugin 2.0.7 (Fiji/ImageJ 1.54p).</i>
274	2026	Distribution (directionality histogram) & Color-coded orientation map	actin cytoskeleton in cancer-associated fibroblasts	Garside et al. (2026)	Advanced Science	10.1002/advs.202522440	<i>Angle distributions were calculated using the OrientationJ Distribution function with a local Gaussian window of $\sigma = 2$ pixels.</i>
275	2026	Coherency map	actin cytoskeleton in tumors	Kaivola et al. (2026)	Nature Materials	10.1038/s41563-025-02473-7	<i>Changes in the orientation (coherency) were assessed using the OrientationJ package in ImageJ software from MIPs of the confocal microscopy images (actin and β-catenin staining)</i>
276	2026	Coherency map + Dominant Direction	actin cytoskeleton, endothelial mechanotransduction	Franz et al. (2026)	Advanced Science	10.1002/advs.202516423	<i>Apparent coherency analysis of HUVECs exposed to different shear stress profiles was performed using the ImageJ 'OrientationJ' plugin and the 'Dominant Direction' tool based on phase contrast time-lapse images.</i>
277	2026	orientation/coherency analysis	bioprinted tissue engineering	Neuhäusler et al. (2026)	Bioactive Materials	10.1016/j.bioactmat.2025.09.027	<i>The line diameter and fiber alignment were subsequently analyzed using fluorescence microscopy and the OrientationJ plugin (FIJI).</i>
278	2026	Coherency map	bone marrow fibrosis in myelofibrosis	Melo et al. (2026)	Hematological Oncology	10.1002/hon.70163	<i>Fiber organization (coherency) and proximity (uniformity) were evaluated with the OrientationJ plugin using the same AOIs.</i>
279	2026	Distribution (directionality histogram)	bone/mineralized tissue	Lin et al. (2026)	npj Regenerative Medicine	10.1038/s41536-025-00448-5	<i>the Riesz Filters method was utilized within the OrientationJ plugin for orientation analysis, and orientation distribution map</i>
280	2026	Directionality (directionality histogram)	bone/mineralized tissue	Zhao et al. (2026)			<i>The directionality of aligned collagen matrix and new bone matrix was analyzed using the "Directionality" and "OrientationJ" plugins in Fiji software.</i>
281	2026	orientation/coherency analysis	cardiac endothelial cells	Brown et al. (2026)	Annals of Biomedical Engineering	10.1007/s10439-026-03973-6	<i>alignment analysis was performed on 10 representative non-overlapping images for each shear condition using ImageJ with the OrientationJ plugin</i>
282	2026	Distribution (directionality histogram) + structure-tensor order parameter	cardiac tissue engineering	Mousavi et al. (2026)	Small	10.1002/sml.202504493	<i>Cell alignment was evaluated using the OrientationJ plugin by generating orientation histograms and calculating the Orientational Order Parameter (OOP).</i>
283	2026	orientation/coherency analysis	cell alignment in bioelectronic tissue	Cheah et al. (2026)	Advanced Healthcare Materials	10.1002/adhm.202502757	<i>Image analysis was performed on FIJI-J using the OrientationJ plugin to determine cell alignment.</i>
284	2026	orientation/coherency analysis	cell membrane cytoskeleton architecture	Bathe-Peters et al. (2026)	npj Microgravity	10.1038/s41526-026-00587-z	<i>Extraction of orientation analysis using OrientationJ for control and μg conditions at the basolateral membrane and a plane through the perinuclear region</i>
285	2026	Coherency map	collagen/ECM in ovarian fibrosis	Aitken et al. (2026)	Molecular Reproduction and Development	10.1002/mrd.70107	<i>images were analyzed using the OrientationJ plugin to generate a coherency map and calculate the directional coherency coefficient</i>
286	2026	Distribution (directionality histogram)	collagen/ECM in tumors	Corsini et al. (2026)	Cell Death & Disease	10.1038/s41419-025-08404-3	<i>we investigated the fibril orientation in ECM in SK-Mel-31 and OVCAR3-derived tumor sections using the OrientationJ plugin of ImageJ, which provide a visual representation of fibril orientation distribution</i>
287	2026	structure tensor & coherency analysis	connective tissue biomechanics	Karunaseelan et al. (2026)	PLOS One	10.1371/journal.pone.0343718	<i>The local orientation and isotropic properties of the fibrous structures were derived from a 2 x 2 symmetric structure tensor generated for each pixel using ImageJ software and the OrientationJ plugin</i>
288	2026	Distribution (directionality histogram)	electrospun fibers, bone tissue engineering	Zhou et al. (2026)	Bioactive Materials	10.1016/j.bioactmat.2026.05.041	<i>We used the OrientationJ Distribution tool to quantify the percentage of different groups of electrospun fibres in each angular orientation.</i>
289	2026	Distribution (directionality histogram)	materials/nanofibers	Vihmar et al. (2026)	Frontiers in Robotics and AI	10.3389/frobt.2026.1736915	<i>Fiber orientation distributions were measured using the ImageJ plugin OrientationJ.</i>
290	2026	orientation/coherency analysis	materials/nanofibers	Marcelin et al. (2026)			<i>Fiber diameter and fiber alignment were calculated from 4 images (2574×) from these locations using the General Image Fiber Tool (GIFT) [26] and OrientationJ plugin in FIJI</i>
291	2026	orientation/coherency analysis	materials/nanofibers	Livirska et al. (2026)			<i>The orientation of the fibers was determined using the OrientationJ plugin [31] to the ImageJ software 1.51w.</i>
292	2026	Distribution (directionality histogram)	materials/nanofibers	Dos et al. (2026)	ACS Polymers Au	10.1021/acspolymersau.6c00010	<i>The orientation distribution was quantified from SEM micrographs and obtained using OrientationJ</i>
293	2026	Fiber orientation analysis	materials/nanofibers	Oosthuysen et al. (2026)	Biomaterials and Biosystems	10.1016/j.bbiosy.2026.100133	<i>Fiber diameter and orientation, and pore size (n=6) were determined from SEM images using open source image processing software Fiji with DiameterJ and OrientationJ plugins.</i>
294	2026	Distribution (directionality histogram)	materials/nanofibers	Trifan et al. (2026)	Materials	10.3390/ma19112375	<i>Fibre orientation was quantified on low-magnification SEM micrographs using the OrientationJ plugin implemented in Fiji.</i>
295	2026	Distribution (directionality histogram)	materials/nanofibers	Liu et al. (2026)	Bioactive Materials	10.1016/j.bioactmat.2025.09.026	<i>the feature "OrientationJ Distribution" was executed with an object size of one pixel and the cubic spline gradient with the parameters of 60 % coherence and 1 % energy.</i>
296	2026	Distribution (directionality histogram)	materials/nanofibers	Tort et al. (2026)	ACS Omega	10.1021/acsomega.5c12433	<i>The orientation and diameter of the fibers were determined using OrientationJ and DiameterJ, respectively.</i>
297	2026	Distribution (directionality histogram)	muscle tissue engineering, cultured meat	Kamel et al. (2026)	Foods	10.3390/foods15101715	<i>The relative frequency (%) of myotube orientation was determined using the OrientationJ plugin (Version 2.0.7)</i>
298	2026	Coherency map	musculoskeletal tissue engineering	Spagnuolo et al. (2026)	Advanced Science	10.1002/advs.202509313	<i>Collagen fibril orientation, dispersion, and coherency were analyzed using the OrientationJ and Directionality plugins in ImageJ.</i>
299	2026	Orientation/coherency analysis	neuronal alignment and nanofibers	Morita et al. (2026)	Stem Cell Reports	10.1016/j.stemcr.2026.102923	<i>The orientation of CLDN11 and nanofibers was quantified using the ImageJ OrientationJ plugin (NIH, Bethesda, MD, USA).</i>
300	2026	Dominant Direction	neuronal microglia alignment	Ferro et al. (2026)	Cell Reports	10.1016/j.celrep.2026.116926	<i>we applied the ImageJ plug-in OrientationJ to assess the orientation of these tubular microglia relative to the dendrites of PYR neurons</i>

#	Year	OrientationJ mode	Field / application	Reference	Journal	DOI	Usage extract
301	2026	Coherency and energy metrics	scleral collagen fibers in glaucoma	Masuda et al. (2026)	Graefe's Archive for Clinical and Experimental Ophthalmology	10.1007/s00417-025-07112-0	Local two-dimensional fiber orientations were quantified using OrientationJ, with coherency (an index of local structural anisotropy) and energy (an index reflecting the magnitude of orientation information) as outcome variables.
302	2026	Coherency map	spindle microtubule organization	Gohel et al. (2026)	Journal of Cell Science	10.1242/jcs.264165	Spindle MT organization was quantified from α -tubulin images using the OrientationJ coherency index in Fiji/ImageJ, which reports local filament alignment (0=random, 1=perfectly ordered)
303	2026	Dominant Direction	stem cells and liver tissue engineering	Xue et al. (2026)	Bioactive Materials	10.1016/j.bioactmat.2026.05.025	Filament orientation was analyzed with the OrientationJ plugin, using the cell axis as 0° reference.
304	2026	orientation/coherency analysis	vascular graft fiber alignment	Fahad et al. (2026)	Bioactive Materials	10.1016/j.bioactmat.2026.03.040	Anisotropy of the fiber alignment was evaluated using the OrientationJ plugin.